

Notes: Bekaert, Harvey, and Lundblad (JFE, 2005)

Does Financial Liberalization Spur Growth?

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1 Big picture

- **Question.** Does opening a country's equity market to foreign investors raise subsequent real economic growth?
- **Core finding.** Equity market liberalization is associated with about a one percentage point increase in annual real per-capita GDP growth.
- **Economic mechanism.** Liberalization can lower the cost of equity capital, improve international risk sharing, relax financing constraints, increase foreign capital availability, and strengthen corporate governance through outside investor pressure.
- **Why the question is not obvious.** Capital-account liberalization studies often find mixed or fragile growth effects. Financial openness could also reduce precautionary saving or fail to allocate capital well if information problems are severe.
- **Main empirical design.** The paper uses a cross-country panel from 1980 to 1997 and adds equity market liberalization variables to standard growth regressions with macroeconomic and demographic controls.
- **Main measurement object.** The baseline variable is an **Official Liberalization** indicator that turns on when formal regulatory changes allow foreign investors to hold domestic equities.
- **Alternative liberalization measures.** The paper also uses **First Sign** dates, which capture the first official opening, ADR introduction, or country fund launch, and a continuous **Liberalization Intensity** measure based on the share of the local market that is investable by foreigners.
- **Capital-account comparison.** The standard IMF capital-account openness dummy does not robustly predict growth, while the finer Quinn measure does better. The equity market liberalization effect survives even when these capital-account measures are included.
- **Endogeneity concern.** Countries may open equity markets when future growth prospects are already improving. The paper addresses this by constructing an external industry-based measure of growth opportunities and showing that the liberalization coefficient barely changes.
- **Reform-bundle concern.** Equity liberalization can coincide with trade reforms, disinflation, exchange-rate reforms, financial development, legal reforms, insider-trading enforcement, and post-crisis reforms. These variables reduce the coefficient somewhat, but do not eliminate it.
- **Heterogeneity.** The growth response is largest in countries with better financial development, stronger institutions, better investor protection, and better investment conditions.
- **Big takeaway.** The paper argues that equity market liberalization is not merely a symbolic component of broad financial opening. It has a distinct and economically large association with growth, although the authors are careful that the evidence is predictive and not a definitive causal experiment.

2 Data and measurement (key objects)

2.1 Outcome variable: real per-capita GDP growth

The dependent variable is growth in real GDP per capita. The main regressions use a five-year forward average growth rate. The paper focuses on annual data from 1980 through 1997.

Interpretation. The five-year horizon is long enough to capture medium-run growth effects, but short enough to use information from a relatively short post-1980 panel with many liberalizations occurring in the late 1980s and early 1990s.

2.2 Country samples

The paper works with several samples because different controls are available for different countries.

- **Sample I:** 95 countries, the largest overall sample, available for the baseline growth regressions and core macro controls.
- **Sample II:** 50 countries, the largest sample that also has detailed financial development variables such as private credit and turnover.
- Smaller samples are used when legal, institutional, or financial variables have limited coverage.

Interpretation. The baseline sample is broad enough to compare fully liberalized, never liberalized, and liberalizing countries. The smaller samples allow richer controls, but at the cost of external validity and power.

2.3 Baseline equity market liberalization variable

The key variable is **Official Liberalization**. It equals one in and after the year when formal regulatory changes allow foreign investors to invest in domestic equity securities. For fully segmented countries, it is always zero. For fully liberalized countries, it is always one.

Interpretation. This variable is a country-year indicator for direct equity market access. It focuses on stock market openness rather than broad capital-account openness.

2.4 Alternative equity market liberalization measures

The paper checks whether the result is sensitive to how liberalization is dated or measured.

- **First Sign.** This turns on at the earliest of the official liberalization date, the first ADR announcement, or the first country fund launch.
- **Liberalization Intensity.** This is the ratio of IFC Investable market capitalization to IFC Global market capitalization. A value of one means that all stocks are available to foreign investors.

Interpretation. First Sign captures indirect access before full regulatory opening. Liberalization Intensity captures the depth of investability rather than a simple zero-one date.

2.5 Capital-account openness measures

The paper distinguishes equity market liberalization from broader capital-account liberalization.

- **IMF capital-account openness.** A zero-one measure based on whether the country reports at least one restriction on capital-account payments.
- **Quinn capital-account openness.** A more graded measure based on the IMF's annual descriptions, scored from closed to open and rescaled to lie between zero and one.

Interpretation. The IMF dummy is coarse. The Quinn variable is more nuanced and therefore has more variation in the intensity of openness.

2.6 Baseline growth controls

The standard controls follow the cross-country growth literature:

- initial real GDP per capita, usually log real GDP per capita in 1980;
- government consumption divided by GDP;
- secondary school enrollment;
- population growth;
- log life expectancy.

Interpretation. These controls proxy for the steady-state level of income and convergence forces. Conditional convergence predicts that, holding steady-state determinants fixed, initially poorer countries should grow faster.

2.7 Reform and institutional variables

To test whether equity liberalization is really proxying for other reforms, the paper adds several groups of variables:

- **Macroeconomic reforms:** trade openness, inflation, and black-market exchange-rate premia.
- **Financial development:** private credit to GDP and equity market turnover.
- **Legal environment:** law and order, legal origin, judicial efficiency, and speed of legal process.
- **Investor protection:** insider trading laws, insider trading prosecutions, anti-director rights, creditor rights, and accounting standards.
- **Banking crises:** crisis-period and post-crisis indicators based on systemic or borderline banking crises.
- **Institutional quality:** ICRG-based institutional and investment-condition measures.

Interpretation. These variables ask whether the measured equity liberalization effect is actually a bundled reform effect. If the liberalization coefficient remains positive after adding them, the equity market opening result is harder to dismiss as pure omitted reform bias.

2.8 Descriptive pattern

Before running full regressions, the paper shows that growth is higher after liberalization. In Table 2, five-year average real GDP growth rises from 1.59% before liberalization to 2.76% after liberalization. Table 3 then shows that the post-liberalization growth difference remains positive in fixed-effect and time-effect regressions.

Interpretation. The raw evidence is not causal, but it motivates the main growth regressions. The central question is whether the approximately one percentage point pattern survives controls, alternative measurement, capital-account variables, and endogeneity checks.

3 Models and empirical specifications

3.1 Five-year average growth rate

The paper defines the five-year average growth rate as

$$y_{i,t+k,k} = \frac{1}{k} \sum_{j=1}^k y_{i,t+j}, \quad (1)$$

where annual real per-capita GDP growth is

$$y_{i,t} = \ln \left(\frac{GDP_{i,t}/POP_{i,t}}{GDP_{i,t-1}/POP_{i,t-1}} \right). \quad (2)$$

Interpretation. The dependent variable is average future growth, not contemporaneous growth. This makes the liberalization indicator a predictor of subsequent medium-run growth.

3.2 Standard conditional-growth regression

A standard cross-country growth regression can be written as

$$y_{i,t+k,k} = -\lambda Q_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t+k,k}, \quad (3)$$

where $Q_{i,t}$ is log per-capita GDP and $X_{i,t}$ contains steady-state determinants.

Interpretation. The coefficient $-\lambda$ captures conditional convergence. Countries with lower initial income should grow faster once human capital, demographics, government size, and health are held fixed.

3.3 Main liberalization regression

The paper's main addition is to include an equity market liberalization variable:

$$y_{i,t+k,k} = \beta Q_{i,1980} + \gamma' X_{i,t} + \alpha Lib_{i,t} + \varepsilon_{i,t+k,k}. \quad (4)$$

Interpretation. The coefficient α is the object of interest. If $\alpha = 0.010$, liberalization predicts about one additional percentage point of annual real per-capita GDP growth over the future five-year window.

3.4 Preliminary fixed-effect specification

Before the full growth regression, the paper estimates a simpler annual panel regression of the form

$$y_{i,t} = \mu_i + \tau_t + \alpha Lib_{i,t} + u_{i,t}, \quad (5)$$

where μ_i are country fixed effects and τ_t are time effects.

Interpretation. Country fixed effects absorb permanent country-level growth differences. Time effects absorb global shocks. The coefficient is therefore identified from growth changes within countries around liberalization relative to the global cycle.

3.5 OLS with non-overlapping windows

The first full-regression approach uses non-overlapping five-year growth averages. Because the available period is short, the paper reports averages across three possible non-overlapping samples: 1981–1995, 1982–1996, and 1983–1997.

Interpretation. This approach is transparent and avoids mechanically overlapping dependent variables, but it throws away some time-series information.

3.6 GMM with overlapping windows

The second approach uses overlapping five-year growth windows and estimates the model with a generalized method of moments procedure. Standard errors correct for cross-sectional heteroskedasticity and for the moving-average structure created by overlapping growth windows.

Interpretation. The overlapping estimator uses more of the panel's timing information. The price is that residuals are mechanically serially correlated, so standard errors must be adjusted.

3.7 Capital-account comparison model

To test whether equity market liberalization simply captures general capital mobility, the paper estimates regressions that include both equity liberalization and capital-account openness:

$$y_{i,t+k,k} = \beta Q_{i,1980} + \gamma' X_{i,t} + \alpha Lib_{i,t}^{Equity} + \delta Lib_{i,t}^{Capital} + \varepsilon_{i,t+k,k}. \quad (6)$$

Interpretation. If α remains positive after including $Lib_{i,t}^{Capital}$, then the equity market channel is not subsumed by broad capital-account openness.

3.8 Growth-opportunity control

The paper constructs an external measure of country-specific growth opportunities using global industry price-earnings ratios and the country's industrial composition:

$$GO_{i,t} = \ln \left(\frac{IPE_t w'_{i,t}}{WDPE_t} \right) - \frac{1}{60} \sum_{s=t-60}^{t-1} \ln \left(\frac{IPE_s w'_{i,s}}{WDPE_s} \right). \quad (7)$$

Here IPE_t is the vector of global industry price-earnings ratios, $w_{i,t}$ is the vector of country-specific industry output weights, and $WDPE_t$ is the world market price-earnings ratio.

Interpretation. This variable is designed to be external to the country's own liberalization choice. If global industry valuations predict that a country's industry mix has strong opportunities, that country may grow faster regardless of liberalization. Including $GO_{i,t}$ directly addresses the concern that countries liberalize right before predictable growth improvements.

3.9 Reform-bundle regression

A generic reform-control specification is

$$y_{i,t+k,k} = \beta Q_{i,1980} + \gamma' X_{i,t} + \alpha Lib_{i,t} + \theta' R_{i,t} + \varepsilon_{i,t+k,k}, \quad (8)$$

where $R_{i,t}$ is a vector of reform proxies such as trade openness, inflation, black-market premia, private credit, turnover, law and order, insider-trading enforcement, and banking-crisis indicators.

Interpretation. The goal is not to estimate a complete reform model. The goal is to ask whether the liberalization coefficient disappears once observable reform channels are included.

3.10 Heterogeneity model

To study why some countries respond more than others, the paper effectively separates the liberalization effect by country characteristics:

$$y_{i,t+k,k} = \beta Q_{i,1980} + \gamma' X_{i,t} + \alpha_F Lib_{i,t}^{Full} + \alpha_L Lib_{i,t} \mathbf{1}\{Z_i \text{ low}\} + \alpha_H Lib_{i,t} \mathbf{1}\{Z_i \text{ high}\} + \varepsilon_{i,t+k,k}. \quad (9)$$

Interpretation. The high-low comparison asks whether liberalization is more growth-enhancing when institutions, investor protections, or financial systems are already strong.

4 Identification and interpretation

4.1 What identifies the main coefficient?

The key coefficient is identified by differences in subsequent growth across country-years with different liberalization status, after controlling for initial GDP and standard growth determinants. In the fixed-effect regressions, it is additionally identified by within-country changes around liberalization and by comparisons to global time effects.

Interpretation. The paper's evidence is a predictive panel-growth design. It is stronger than a simple before-after comparison, but weaker than a randomized or natural-experiment design.

4.2 Why the result is not just the global business cycle

Many countries liberalized around similar periods. If the world economy accelerated at the same time, the liberalization coefficient could pick up global growth. The paper addresses this by adding time effects in preliminary regressions and by adding OECD growth and world real interest rates in the main GMM regressions.

Interpretation. The liberalization effect remains when world-cycle controls are included. In fact, the coefficient does not fall when those controls enter.

4.3 Why the result is not just capital-account openness

Broad capital-account openness is conceptually different from equity market access. A country can have capital controls yet still allow some equity inflows, or it can report openness in a way that does not precisely capture equity market investability.

Interpretation. The IMF capital-account dummy is fragile and generally insignificant. The Quinn measure is more successful, but the equity market coefficient remains positive and significant when included jointly.

4.4 Why growth opportunities are important

The main reverse-causality concern is that governments open equity markets when growth opportunities are already improving. If that is true, liberalization would predict growth even if it had no effect.

The paper's industry-based growth-opportunity measure helps because it uses global industry price-earnings ratios interacted with the country's industrial structure. This is meant to proxy for externally generated opportunities facing each country.

Interpretation. The growth-opportunity control predicts growth, but the official liberalization

coefficient changes very little: from 0.97% in the baseline GMM to 0.92% when growth opportunities are added.

4.5 Why simultaneous reforms matter

Equity liberalization may be part of a broader reform package. If so, the coefficient on $Lib_{i,t}$ could capture trade reforms, macro stabilization, better banking, legal improvements, or post-crisis restructuring.

Interpretation. The paper adds observable proxies for these reforms. The liberalization coefficient falls in some cases, especially with macroeconomic controls, but remains positive and statistically significant.

4.6 Why institutions change the effect

The paper finds larger growth responses in countries with better financial development, institutions, legal quality, investor protection, and investment conditions.

Interpretation. Liberalization is not a mechanical input. Foreign capital and market access are more useful when domestic institutions can allocate capital well, protect investors, and enforce contracts.

4.7 Remaining limitation

The authors explicitly state that the regressions reveal association rather than definitive causality. There is no perfect instrument for the liberalization decision, and the timing of liberalization may still be related to unobserved policy and political changes.

Interpretation. The strongest reading is that the paper provides robust cross-country predictive evidence that equity market liberalization is followed by faster growth, not a clean randomized estimate of the causal effect.

5 Tables and figures

5.1 Table 1: Variable definitions

Read: Table 1 defines the liberalization variables, capital-account variables, banking-crisis variables, insider-trading variables, macro controls, financial development variables, legal variables, and institutional variables.

Economic message. The paper's contribution depends heavily on measurement. The equity market liberalization measures are more specific than broad capital-account measures, which helps explain why the paper finds a stronger and less fragile growth relation than much of the capital-account liberalization literature.

5.2 Table 2: Summary statistics around liberalization

Read: Table 2 compares pre- and post-liberalization averages. Five-year real GDP growth rises from 1.59% before liberalization to 2.76% after liberalization. The table also shows changes in education, population growth, life expectancy, growth opportunities, black-market premia, legal variables, and institutional variables.

Economic message. Liberalization episodes coincide with broad improvements in economic and institutional conditions. This is exactly why the later regressions must control for reform bundles and growth opportunities.

Table 2
Summary statistics. We explore the three, five, and seven-year averages of the growth rate of real per capita gross domestic product (GDP) and the five-year averages of the other variables employed in the paper (and summarized in Table 1) before and after the equity market liberalization (including the liberalization year in the after period). For some countries, we do not have a full three, five, or seven years available given the timing of the liberalization, so we simply take the available years in the average. For all variables, unless otherwise stated, the summary statistics reflect data for 95 countries from 1980 to 1997. Official Liberalization means that the equity market is liberalized. Fully liberalized denotes countries that are fully liberalized throughout our sample. Never liberalized denotes countries that never undergo financial liberalization. ICRG is the International Country Risk Guide. Statistical significance is denoted by * for 10%, ** for 5%, and *** for 1%. NA denotes variables for which the test is not available.

Variable	Pre-liberalization	Post-liberalization	Never liberalized	Fully liberalized
Real GDP growth (three-year)	0.0160	0.0265**	-0.0016	0.0201***
Real GDP growth (five-year)	0.0139	0.0276***		
Real GDP growth (seven-year)	0.0153	0.0264***		
Government/GDP	0.1378	0.1326	0.1581	0.1885***
Enrollment	0.5573	0.6115**	0.5439	0.9974***
Population growth	0.0203	0.0169**	0.0225	0.0666***
Life expectancy	65.7	67.7***	56.9	75.3***
Growth opportunity	-0.0001	0.0076***	-0.0012	-0.0016
Trade/GDP	0.6229	0.6383	0.6070	0.8429***
Log(1 + inflation) (Latin)	0.1890	0.1411	0.0596	NA
Log(1 + inflation) (not Latin)	0.0993	0.0857	0.0934	0.0401***
Log(1 + black market premium)	0.1999	0.0724***	0.2211	0.0007***
Fiscal Deficit (28 countries)	0.0606	0.0333***	NA	0.0307
Private credit/GDP	0.3851	0.4263	0.2286	0.8086***
Turnover (30 countries)	0.1814	0.2664	NA	0.4938
Banking crisis (systematic)	0.3243	0.2941	0.5300	0.1131**
Banking crisis (systematic and herds)	0.5243	0.5794	0.4390	0.3091
Law and order (75 countries)	0.4875	0.6065***	0.4472	0.9516***
Insider trading law	0.4205	0.7241***	0.0836	0.6540***
Insider trading prosecution	0.0667	0.1149*	NA	0.4325
Judicial efficiency (47 countries)			NA	0.9456
Speed of process (checks + evictions) (49 countries)			363.4	408.3
Quality of institutions (73 countries)	0.5273	0.6033***	0.4158	0.9333***
ICRG economic index (75 countries)	0.5095	0.6765***	0.5009	0.7845
Investment profile (75 countries)	0.4660	0.5311**	0.4640	0.6494***
Anti-director rights (47 countries)			NA	0.4902
Creditor rights (45 countries)			NA	0.4653
Accounting standards (39 countries)			NA	0.6950

Table 3
Preliminary analysis of the impact of liberalization. For all estimates, the dependent variable is the one-year average growth rate of real per capita gross domestic product (GDP). Regressions include time effects, fixed effects, or both, as indicated (not reported in the interest of space); no other controls are included. In Panel A, we focus on equity market liberalization across the 40 countries that liberalize in our sample. The Official Liberalization variable takes a value of one when the equity market is liberalized, and zero otherwise. We consider an additional regression that includes China (41 countries). The First Sign liberalization indicator takes the value of one after the first of the following events: the Official Liberalization date, the introduction of American Depository Receipts, or the introduction of a country fund. The Liberalization Intensity measure is the ratio of the market capitalizations for the International Finance Corporation's investable to global indices.

In Panel B, we consider more general measures of capital account openness. The International Monetary Fund capital account openness indicator takes on value of zero if the country has at least one reported capital account restriction. The Quinn capital account liberalization indicator takes a value between one and zero depending upon the intensity of the reported capital account liberalization or openness; these regressions include 76 countries. For both measures, we perform regressions for the same 40 liberalizing countries for comparison, as well as for the full set of countries for which the measures are available.

	Estimate	Standard Error	Adjusted R ²
Panel A: Equity market liberalization			
Official Liberalization indicator (40 countries)			
Fixed effects	0.0124	0.0032	0.208
Time effects	0.0202	0.0048	0.052
Fixed and time effects	0.0105	0.0053	0.229
Official Liberalization indicator plus China (41)			
Fixed effects	0.0128	0.0031	0.251
Time effects	0.0210	0.0049	0.048
Fixed and time effects	0.0117	0.0053	0.270
First Sign indicator (40)			
Fixed effects	0.0129	0.0033	0.208
Time effects	0.0185	0.0041	0.055
Fixed and time effects	0.0080	0.0050	0.228
Liberalization Intensity (40)			
Fixed effects	0.0205	0.0051	0.209
Time effects	0.0157	0.0064	0.033
Fixed and time effects	0.0151	0.0064	0.231
Panel B: Capital account liberalization			
IMF capital account openness indicator (40)			
Fixed effects	0.0036	0.0065	0.190
Time effects	0.0037	0.0043	0.029
Fixed and time effects	0.0017	0.0065	0.224
IMF capital account openness indicator (95)			
Fixed effects	0.0041	0.0051	0.110
Time effects	0.0071	0.0029	0.024
Fixed and time effects	-0.0017	0.0053	0.133
Quinn capital account openness indicator (37)			
Fixed effects	0.0154	0.0192	0.169
Time effects	0.0218	0.0086	0.030
Fixed and time effects	-0.0016	0.0203	0.196

Table 3 (continued)

	Estimate	Standard Error	Adjusted R ²
Quinn capital account openness indicator (76)			
Fixed effects	0.0122	0.0123	0.143
Time effects	0.0193	0.0047	0.053
Fixed and time effects	0.0019	0.0129	0.167

Table 4
The impact of liberalization in pooled ordinary least squares (OLS) growth regressions. For all estimates, the dependent variable is the five-year nonoverlapping average growth rate of real per capita gross domestic product (GDP). Log(GDP) is the log real per capita GDP level in 1980. Govt/GDP is the ratio of government consumption to GDP; enrollment is the secondary school enrollment ratio; population growth is the growth rate of total population; Log(life) is the log life expectancy of the total population. In Panel A, the Official Liberalization variable takes a value of one when the equity market is liberalized, and zero otherwise; these regressions cover 95 countries.

In Panel B, the International Monetary Fund (IMF) capital account openness indicator takes on value of zero if the country has at least one reported capital account restriction; these regressions cover 95 countries. In Panel C, the Quinn capital account liberalization indicator takes a value between one and zero depending upon the intensity of the reported capital account liberalization or openness; these regressions include 76 countries. We first consider each control variable separately, then all together. For each case, we report the simple average of three coefficients (with standard errors and adjusted R²s) associated with separate pooled OLS regressions (over 1981-1995, 1982-1996, and 1983-1997) for which the dependent variable is three nonoverlapping five-year GDP average growth rates. That is, each pooled OLS regression has three time-series observations with no overlap; we conduct each regression separately and then average the resulting coefficients. OLS standard errors are below each estimate in parentheses; for the last entry of each panel, we also include restricted seemingly unrelated regression (SUR) standard errors (all off-diagonal elements are assumed to be equal) as a robustness check.

Constant	Initial log(GDP)	Govt/GDP	Secondary school enrollment	Population growth	Log(life)	Official Liberalization indicator	IMF capital account openness	Quinn capital account openness	Adjusted R ²
Panel A: Official Liberalization (95 countries)									
0.0048 (0.0021)						0.0181 (0.0029)			0.082
0.0020 (0.0104)		0.0004 (0.0015)				0.0173 (0.0048)			0.079
0.0072 (0.0052)			-0.0152 (0.0332)			0.0182 (0.0030)			0.081
-0.0011 (0.0035)				0.0145 (0.0073)			0.0119 (0.0048)		0.094
0.0135 (0.0041)					-0.3568 (0.1479)		0.0127 (0.0038)		0.106

5.3 Table 3: Preliminary fixed-effect and time-effect regressions

Read: Table 3 regresses one-year real per-capita GDP growth on liberalization variables with fixed effects, time effects, or both. For the official liberalization indicator, the fixed-and-time-effect coefficient is 0.0105 in the 40-country liberalizing sample. The First Sign coefficient is 0.0080, and the Liberalization Intensity coefficient is 0.0151.

Economic message. Even before adding the full Barro-style control set, the post-liberalization growth effect is around one percentage point and survives global time effects.

5.4 Table 4: Pooled OLS growth regressions

Read: Table 4 estimates non-overlapping five-year growth regressions. In the full specification with the standard controls, the official liberalization coefficient is 0.0120 with an OLS standard error of 0.0044 and a restricted-SUR standard error of 0.0025.

Economic message. The coefficient implies about 1.20 percentage points higher annual growth after liberalization. The effect is large and statistically meaningful even in a standard conditional-growth regression.

Table 5
Equity market and capital account liberalization. The dependent variable is the overlapping five-year average growth rate of real per capita gross domestic product (GDP). In addition to the control variables, the model includes coefficients for International Finance Corporation (IFC) investment and the index of the market is liberalized, and zero otherwise. The First Sign Domestic product index takes the value of one after the first of the following events: the Official Liberalization date, the introduction of an American Depository Receipt, or the introduction of a country fund. The Liberalization Intensity is the ratio of the average of the coefficients for International Finance Corporation (IFC) investment and the index of the International Monetary Fund (IMF) capital account liberalization index takes on a value of zero if the country has at most one reported capital account restriction; these regressions cover 95 countries. In Panel A, the intensity of the reported capital account liberalization or openness; these regressions cover 76 countries. All standard errors (in parentheses) provide a correction for cross-sectional heteroskedasticity

	(1)	(2)	(3)	(4)
<i>Panel A: Full sample (95 countries)</i>				
Constant	-0.3277 (0.0286)	-0.3240 (0.0278)	-0.3370 (0.0288)	-0.3267 (0.0287)
Initial logGDP	-0.0082 (0.0010)	-0.0082 (0.0010)	-0.0086 (0.0011)	-0.0082 (0.0011)
Gov/GDP	-0.0144 (0.0131)	-0.0102 (0.0122)	-0.0135 (0.0131)	-0.0142 (0.0133)
Secondary school enrollment	0.0004 (0.0004)	-0.0019 (0.0045)	-0.0003 (0.0049)	0.0006 (0.0049)
Population growth	-0.011 (0.0078)	-0.1474 (0.0753)	-0.1922 (0.0776)	-0.1935 (0.0783)
LogGDP	0.0975 (0.0076)	0.0966 (0.0074)	0.1007 (0.0078)	0.0974 (0.0077)
Official Liberalization indicator	0.0094 (0.0020)			0.0092 (0.0021)
First Sign Liberalization indicator		0.0122 (0.0020)		
Liberalization Intensity			0.0107 (0.0023)	
IMF capital account openness indicator				0.0010 (0.0017)
Adjusted R ²	0.207	0.215	0.206	0.207
	(1)	(2)	(3)	(4)
<i>Panel B: Quotient sample (76 countries)</i>				
Constant	-0.2862 (0.0150)	-0.2808 (0.0144)	-0.3072 (0.0144)	-0.2847 (0.0149)
Initial logGDP	-0.0011 (0.0001)	-0.0161 (0.0001)	-0.0110 (0.0001)	-0.0104 (0.0001)

Table 5 (continued)

	(1)	(2)	(3)	(4)	(5)
Gov/GDP	-0.0152 (0.0162)	-0.0305 (0.0155)	-0.0220 (0.0160)	-0.0334 (0.0145)	-0.0377 (0.0161)
Secondary school enrollment	0.0026 (0.0050)	-0.0007 (0.0049)	0.0008 (0.0050)	0.0024 (0.0052)	0.0037 (0.0054)
Population growth	-0.4241 (0.1056)	-0.4241 (0.1036)	-0.4313 (0.1053)	-0.4424 (0.1088)	-0.4550 (0.1107)
Log(life)	0.0947 (0.0099)	0.0933 (0.0087)	0.0991 (0.0088)	0.0988 (0.0089)	0.0966 (0.0085)
Official Liberalization indicator	0.0120 (0.0022)			0.0115 (0.0022)	0.0077 (0.0023)
First Sign liberalization indicator		0.0149 (0.0021)			
Liberalization Intensity			0.0147 (0.0025)		
IMF capital account openness indicator				0.0020 (0.0017)	
Quinn capital account openness indicator					0.0179 (0.0040)
Adjusted R ²	0.270	0.286	0.271	0.270	0.284

Table 4
(continued)

Cluster	Initial Log(GDP)	Secondary sector growth	Population growth	Log(HDI)	Official language indicator	IMR (per 1,000 live births)	Quinn index (1996)	Adjusted R ²
-1.199	-0.013	-0.001	-0.007	-0.009	0.048	0.814	0.117	
CE (n = 13)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	0.10
-0.391	-0.004	-0.007	-0.009	-0.006	0.015	0.32	0.149	
CE (n = 13)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	0.17
-0.005	0.006	0.013	0.016	0.047	0.139	0.844	0.001	
Revised CE market entry	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.141	-0.009	-0.014	-0.009	-0.009	0.029	0.803	0.014	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.014
-0.009	-0.009	-0.009	-0.009	-0.009	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	0.001
-0.001	-0.001	-0.001	-0.001	-0.001	0.017	0.001	0.001	
CE (n = 13)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)</			

5.5 Table 5: GMM estimates with overlapping growth windows

Read: Table 5 is the main benchmark table. In the full 95-country sample, the official liberalization coefficient is 0.0097. The First Sign coefficient is 0.0122. The Liberalization Intensity coefficient is 0.0107. When the IMF capital-account openness variable is added, the official liberalization coefficient remains 0.0094 while the IMF coefficient is only 0.0010.

In the 76-country Quinn sample, the official liberalization coefficient is 0.0120 alone. When Quinn openness is included, the equity coefficient falls to 0.0077 but remains statistically strong, and the Quinn coefficient is 0.0179.

Economic message. The main result is not sensitive to liberalization dating. It is also not explained away by capital-account openness.

5.6 Table 6: Regional effects, world cycle, and growth opportunities

Table 6

Analysis of the liberalization effect. Samples 1 and II refer to samples of 95 and 50 countries, respectively. The dependent variable is the overlapping five-year average growth rate of real per capita gross domestic product (GDP). In addition to the control variables, we report the coefficient on the Official Liberalization indicator that takes a value of one when the equity market is liberalized, and zero otherwise. In Panel A, real interest rate is an indicator that takes the value of one if the country is in Latin America. In Panel B, the world real interest rate is the contemporaneous GDP-weighted real interest rate for the G-7 countries. ODG-GDP growth is the five-year average real GDP growth of the organization for Economic Cooperation and Development countries. In Panel C, the variable $\Delta \ln \text{GDP}$ is the five-year average real GDP growth of the country. All variables are in logarithmic form. The standard errors (in parentheses) provide a correction for the cross-sectional heteroskedasticity and account for the overlapping nature of the data.

	Panel A: regional influences		Panel B: world growth and real interest rates		Panel C: growth opportunities	
	Sample I	Sample II	Sample I	Sample II	Sample I	Sample II
Constant	-0.3291 (0.0291)	-0.2793 (0.0272)	-0.3323 (0.0279)	-0.2954 (0.0318)	-0.3352 (0.0409)	-0.2679 (0.0449)
Initial log(GDP)	-0.0082 (0.0010)	-0.0106 (0.0012)	-0.0103 (0.0010)	-0.0114 (0.0012)	-0.0204 (0.0010)	-0.0107 (0.0012)
Gov/GDP	-0.0150 (0.0132)	-0.0705 (0.0166)	-0.0154 (0.0126)	-0.0700 (0.0157)	-0.0115 (0.0128)	-0.0661 (0.0160)
Secondary school enrollment	0.0000 (0.0000)	0.0028 (0.0055)	0.0003 (0.0045)	0.0000 (0.0050)	0.0001 (0.0053)	0.0047 (0.0044)
Population growth	-0.1905 (0.0777)	-0.1885 (0.1228)	-0.1845 (0.0729)	-0.3985 (0.1050)	-0.2053 (0.0768)	-0.4607 (0.1170)
Log(GDP)	0.0980 (0.0777)	0.0937 (0.0126)	0.0998 (0.0073)	0.0974 (0.0021)	0.0974 (0.0097)	0.0909 (0.0124)
Official Liberalization index			0.0108 (0.0019)	0.0112 (0.0021)	0.0092 (0.0021)	0.0087 (0.0021)

Official Liberalization indicator (Latin)	0.0065 (0.0041)	0.0052 (0.0051)		
Official Liberalization indicator (not Latin)	0.0106 (0.0023)	0.0098 (0.0023)		
OECD GDP growth (contemporaneous)		0.5049 (0.0046)	0.6552 (0.0942)	
World real interest rate (contemporaneous)		-0.2240 (0.0670)	-0.1714 (0.0735)	
Growth opportunities			0.0106 (0.0030)	0.0122 (0.0030)
Adjusted R^2	0.207	0.225	0.231	0.209

Read: Table 6 adds regional interactions, world growth and real interest-rate variables, and the external growth-opportunity measure. The liberalization effect is 0.0092 in Sample I after adding growth opportunities, close to the baseline 0.0097.

Economic message. The coefficient is not driven by Latin America, not driven by the world business cycle, and not eliminated by the paper’s best proxy for exogenous country-specific growth opportunities.

5.7 Table 7: Reform environment

Table 7 (continued)

	Panel A: macroeconomic reforms		Panel B: financial development		Panel C: law and order		Panel D: insider trading	
	Sample I	Sample II	Sample I	Sample II	Sample I	Sample II	Sample I	Sample II
Log(GDP)	0.0079 (0.0075)	0.0170 (0.0171)	0.0057 (0.0076)	0.0028 (0.0111)	0.0057 (0.0090)	0.0051 (0.0111)	0.0067 (0.0075)	0.0076 (0.0123)
Official Liberalization indicator	0.0074 (0.0016)	0.0064 (0.0051)	0.0077 (0.0009)	0.0009 (0.0011)	0.0070 (0.0052)	0.0070 (0.0050)	0.0068 (0.0050)	0.0071 (0.0051)
Trade	0.0006 (0.0004)	0.0008 (0.0007)						
Log(1 + Inflation) (Late)	-0.0006 (0.0023)	0.0008 (0.0025)						
Log(1 + Inflation) (pre Late)	0.0002 (0.0002)	0.0027 (0.0071)						
Log(1 + Stock market premium)	-0.0002 (0.0010)	-0.0002 (0.0012)						
Private credit			0.0137 (0.0011)	0.0004 (0.0012)				
Turnover			-0.0001 (0.0011)	0.0029 (0.0031)				
ICRG law and order				-0.0001 (0.0007)	0.0029 (0.0009)			
Insider trading law					0.0001 (0.0003)	-0.0001 (0.0003)		
Insider trading prosecution					0.0012 (0.0024)	0.0015 (0.0020)		
Adjusted R ²	0.20	0.26	0.20	0.20	0.20	0.21	0.20	0.25

Panel E: banking crises

Sample I	Sample II	Sample I	Sample II	Sample I	Sample II	Sample I	Sample II
Constant	-0.3047 (0.0201)	-0.2402 (0.0440)	-0.3037 (0.0261)	-0.2387 (0.0485)	-0.3750 (0.0270)	-0.2625 (0.0470)	-0.3348 (0.0280)
Initial log(GDP)	-0.0040 (0.0009)	-0.0105 (0.0010)	-0.0068 (0.0010)	-0.0107 (0.0010)	-0.0070 (0.0009)	-0.0104 (0.0010)	-0.0107 (0.0010)
GDP	-0.0121 (0.0120)	-0.0745 (0.0130)	-0.0178 (0.0131)	-0.0652 (0.0130)	-0.0648 (0.0129)	-0.0127 (0.0129)	-0.0111 (0.0129)
Secondary school enrollment	0.0003 (0.0000)	0.0022 (0.0001)	0.0012 (0.0001)	0.0007 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0000 (0.0000)
Population growth	-0.1094 (0.0170)	-0.4406 (0.1010)	-0.1034 (0.1174)	-0.4108 (0.1175)	-0.1051 (0.1175)	-0.4004 (0.1181)	-0.1020 (0.1190)
Log(RG)	0.0012 (0.0075)	0.0001 (0.0110)	0.0004 (0.0075)	0.0001 (0.0112)	0.0001 (0.0076)	0.0001 (0.0112)	0.0001 (0.0077)
Official Liberalization indicator	0.0004 (0.0011)	0.0004 (0.0011)	0.0001 (0.0011)	0.0007 (0.0012)	0.0001 (0.0011)	0.0001 (0.0011)	0.0001 (0.0011)
Dating systemic crisis	-0.0012 (0.0014)	-0.0001 (0.0011)					
Dating systemic and headline crisis	-0.0012 (0.0011)	-0.0001 (0.0011)					
Post systemic crisis			0.0009 (0.0010)	0.0022 (0.0013)			
Post systemic and headline crisis			0.0006 (0.0010)	0.0002 (0.0013)			
Adjusted R ²	0.218	0.240	0.225	0.250	0.211	0.223	0.212

Read: Table 7 adds macroeconomic reforms, financial development, legal environment, insider-trading laws and enforcement, and banking-crisis indicators. With macroeconomic reform controls, the official liberalization coefficient falls to 0.0074 in Sample I and 0.0066 in Sample II, but remains significant. With financial development controls, it is 0.0077 and 0.0069. With crisis controls, it remains around 0.008–0.010 depending on specification.

Economic message. Observable reform proxies explain part of the liberalization effect, but not all of it. The evidence suggests that equity market liberalization is related to, but not fully subsumed by, broader reform packages.

5.8 Table 8: Why the growth effect differs across countries

Read: Table 8 splits the liberalization effect by financial development, legal environment, institutional quality, and investment conditions. For private credit, the liberalization effect is 0.0048 at low levels and 0.0105 at high levels. For turnover, it is 0.0017 at low levels and 0.0094 at high levels. For institutional quality, it is 0.0045 at low levels and 0.0129 at high levels.

Economic message. Liberalization works best where domestic financial and institutional infrastructure is already relatively strong. Openness alone is not enough; the domestic system must be able to absorb and allocate foreign capital productively.

Table 8

Why does the growth effect from liberalizations differ across countries? For each interaction variable, we separately conduct regressions that have the five-year average growth rate of real per capita gross domestic product (GDP) as the dependent variable. We include in the regressions the same control variables as presented in Table 4. We also separate the liberalization effect for fully liberalized and liberalizing countries. For liberalizing countries, we estimate interaction effects with the financial development, legal, and investment condition variables. We report the associated impact on GDP growth for a liberalizing country for a low level (below the median of the associated interaction variable for liberalizing countries) and for a liberalizing country at a high level (above the median of the associated interaction variable for liberalizing countries).

We provide the significance of a Wald test, for which the null hypothesis is that the high-low effects are equivalent. We also report the statistical significance of the interaction coefficient; statistical significance is denoted by a * for 10%, ** for 5%, and *** for 1%. Significance levels are based on standard errors that correct for cross-sectional heteroskedasticity and account for the overlapping nature of the data.

The financial development variables are the ratio of private credit to GDP and equity market turnover. The legal environment variables are legal origin (English, French, or other), judicial efficiency, and the combined speed of the process to resolve a bounced check or tenant eviction (longer duration implies a lower speed). The International Country Risk Guide (ICRG) political risk quality of institutions subcomponent is the sum of the following ICRG subcomponents: corruption, law and order, and bureaucratic quality, detailed in Table 1. For all interaction indices, larger values denote improvements. ICRGE is the ICRG economic risk indicator. The investment conditions variables are a measure of economic risk, the investment profile, anti-director (minority shareholders) rights, creditor rights, and accounting standards. The number of countries for which the interaction variable is available is also provided. Finally, some of the variables are available as time series, while others are only available in the cross section; we denote this in the time-series available column.

Impact on growth resulting from liberalization	Fully liberalized	From low level of variable	From high level of variable	Direct effect of interaction variable	Number of countries	Time-series available
Financial development						
Private credit	0.0084**	0.0048	0.0105***	0.0116***	95	Yes
Turnover	0.0134***	0.0017	0.0094***	0.0152***	50	Yes
Legal environment						
French versus English law	0.0072***	0.0068	0.0124**		95	No
Other versus English law	0.0072***	0.0097	0.0124		95	No
Judicial efficiency	0.0105***	0.0069	0.0099	0.0057	47	No
Speed of process (combined)	0.0065	0.0029	0.0084	-0.0002	69	No
Quality of institutions						
ICRG quality of institutions	0.0098***	0.0045	0.0129***	-0.0003	75	Yes
Investment conditions or protection						
ICRGE	0.0049	0.0071	0.0075	0.0096***	75	Yes
Investment profile	0.0060	0.0019	0.0083***	0.0210***	75	Yes
Anti-director rights	0.0117***	0.0018	0.0099**	0.0064***	47	No
Creditor rights	0.0102***	0.0035	0.0089	0.0190***	45	No
Accounting standards	0.0094**	0.0004	0.0110***	0.0058	39	No

5.9 Key numerical map

Evidence	Main magnitude	Interpretation
Table 2 before-after	1.59% to 2.76%	Raw five-year growth rises after liberalization.
Table 3 fixed and time effects	0.0105	Post-liberalization annual growth is about one point higher even after country and time effects.
Table 4 full pooled OLS	0.0120	Standard growth regression gives a 1.20 point effect.
Table 5 benchmark GMM	0.0097	Main overlapping-window estimate is about one point.
Table 5 with IMF openness	0.0094	IMF capital-account openness does not subsume equity liberalization.
Table 5 with Quinn openness	0.0077	More nuanced capital-account openness matters, but equity liberalization remains significant.
Table 6 with growth opportunities	0.0092	Endogenous timing based on observable opportunities is unlikely to fully explain the result.
Table 7 with macro reforms	0.0074	Reform controls reduce but do not eliminate the effect.
Table 8 institutional quality	0.0045 vs. 0.0129	High-quality institutions are associated with much larger liberalization gains.

6 Short summary

- Equity market liberalization predicts about one extra percentage point of annual real per-capita GDP growth.
- The effect appears in before-after comparisons, fixed-effect regressions, standard OLS growth regressions, and overlapping-window GMM regressions.
- The result is robust to alternative liberalization measures: Official Liberalization, First Sign, and Liberalization Intensity.
- The result is not driven by the IMF capital-account openness dummy and survives the more nuanced Quinn openness measure.
- The result survives controls for regional effects, world growth, world real interest rates, and an external measure of growth opportunities.
- Observable reform proxies reduce the coefficient somewhat but do not eliminate it.
- Countries with better financial development, institutional quality, investor protection, and investment conditions experience larger growth responses.
- The evidence is not a clean causal experiment. The paper's contribution is robust predictive evidence that equity market opening is followed by faster growth.

7 Conclusion

Bekaert, Harvey, and Lundblad show that equity market liberalization is followed by economically large and statistically robust growth improvements. The benchmark magnitude is roughly one percentage point of annual real per-capita GDP growth, which is large in the growth literature.

The central reason the paper matters is its focus on equity market liberalization rather than a broad capital-account openness dummy. A direct measure of foreign access to domestic equities produces a clearer result than the coarse IMF capital-account variable. The more nuanced Quinn capital-account measure also predicts growth, but it does not eliminate the equity liberalization effect.

The paper is also careful about interpretation. Liberalization is not randomly assigned, and the authors do not claim a definitive causal experiment. Instead, they build credibility through many checks: alternative liberalization definitions, fixed and time effects, OLS and GMM estimators, world-cycle controls, growth-opportunity controls, reform controls, and heterogeneity analysis.

The final economic message is conditional. Financial liberalization is most valuable when countries have the domestic infrastructure to use it well. Better financial development, stronger institutions, better investor protection, and better investment conditions all magnify the growth response. Equity market opening can spur growth, but the benefits are larger when openness is matched by institutional quality.